Examining the Association Between Grade 11 ELA and Mathematics CAASPP Scores in California Schools*

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I. Abstract

This study examines whether Grade 11 mathematics achievement is associated with English Language Arts/Literacy achievement across California schools. Using 2023–24 California Assessment of Student Performance and Progress (CAASPP) data, I fit a simple linear regression model predicting mean ELA/Literacy scale scores from mean mathematics scale scores (California Department of Education 2024b). The fitted model indicates a strong positive association: a one point increase in a school's mean mathematics score is associated with an estimated 0.876 point increase in its mean ELA/Literacy score. The estimated intercept is 355.4, and the model explains 87.90% of the variability in ELA/Literacy scores ($R^2 = 0.8790$). The association is statistically significant ($t = 108.74$, $p < 2.2 \times 10^{-16}$), with a residual standard error of 25.96. These results contradict the initial expectation that stronger performance in one subject might be associated with weaker performance in the other. However, schools are nested within districts, which suggests the independence assumption may not hold, and diagnostic plots suggest that the linear model may not fully capture the relationship. Therefore, the findings should be interpreted as a school-level association between CAASPP Math and ELA scores.

II. Introduction

A common stereotype is that students who perform well in mathematics are less proficient in writing and language arts, while students who excel in language arts may struggle with mathematics. Although mathematics and English Language Arts/Literacy (ELA) require different knowledge and skills, success in both subjects may also reflect broader academic

*Github link for project: Math 261A Project Link

factors, such as learning opportunities, school resources, and student engagement. Ideally, this association would be examined using student-level data. However, publicly available CAASPP data are reported only at the school level or higher. Therefore, this project examines whether there is an association between 11th-grade Math and ELA CAASPP mean scale scores across California schools. The results describe the relationship between the two subjects for California schools and should not be interpreted for individual students.

To investigate this question, I use data from the 2023–24 California Assessment of Student Performance and Progress (CAASPP) Smarter Balanced Summative Assessments (California Department of Education 2024b). The analysis focuses on California schools and uses the CAASPP mean scale scores for Math and English Language Arts/Literacy to represent proficiency in both subjects. I use simple linear regression to examine whether mathematics proficiency is linearly associated with English Language Arts/Literacy proficiency. The predictor is math mean scale score, and the response is the corresponding mean scale score for English Language Arts/Literacy. If schools with higher mathematics proficiency also tend to have higher English Language Arts/Literacy proficiency, this may suggest that the two outcomes are related at the school level.

The analysis found a strong positive association between mathematics and ELA/Literacy proficiency across California schools, contrary to the initial expectation that stronger performance in one subject might be associated with weaker performance in the other. However, schools are nested within districts, so errors might not be independent, and diagnostic plots suggested that a simple linear model may not fully capture the relationship. Therefore, the results provide evidence of an association but should be interpreted cautiously, and further analysis is needed to determine whether a linear model is appropriate.

In this report, Section III introduces the data and important variables. Section IV explains the methodology and model assumptions. Section V presents the results and relevant plots. Section VI discusses the findings and the limitations of the selected method.

III. Data

The analysis uses the 2023–24 CAASPP statewide research file (California Department of Education 2024b) and the CAASPP research-file record definitions for the variable names and field meanings (California Department of Education 2024a).

The 2023–24 California Assessment of Student Performance and Progress (CAASPP) Smarter Balanced Summative Assessments measure student achievement in English Language Arts/Literacy (ELA) and mathematics for students in Grades 3–8 and Grade 11. The data are publicly available from the California Department of Education and are reported for public and charter schools serving grade 3 to grade 11, districts, counties, and the state.

The observational unit for this analysis is a California school. The original research file contains aggregate results organized by educational entity, student group, grade, and subject. I

restricted the data to school-level records (`Type ID = 07`), all students (`Student Group ID = 001`), only juniors (`Grade = 11`), and the 2023–24 testing year. Because the data are aggregated at the school level, the analysis examines whether schools with higher mean Math scores also tend to have higher mean ELA scores. It cannot determine whether individual students who perform well in Math also perform well in ELA.

The primary variables are the test identifier, the number of students tested with valid scores, and the mean scale score. The test identifier distinguishes English Language Arts/Literacy (`Test ID = 1`) from mathematics (`Test ID = 2`). The number of students tested with scores represents the number of tested students who received valid assessment scores. The response variable is the mean ELA/Literacy scale score, while the predictor variable is the mean mathematics scale score, both of which are from 0-2900. Both mean scale scores measure the average scale score of students in a school for the selected subject and grade. It is calculated by taking the sum of scale scores for students with valid scores in the selected school divided by the Number of Tests with Valid Scores. The dataset also includes contextual variables, such as county code, district code, district name, school code, and school name, which identify each school and its location.

The data were transformed before analysis because ELA and mathematics results were initially stored in separate rows. I selected the relevant school, subject, student group, grade, number of students tested, and mean score variables for ELA and Math. I converted numeric fields to numeric format to ensure the column data types are correct. I then matched the ELA and mathematics records by school so that each row in the final analysis dataset contained both the ELA and mathematics mean scale scores. Schools without valid results for either subject were excluded. After filtering, 512 schools were excluded and 1630 schools remained.

The CAASPP research file provides several measures of achievement like the percentages of students who exceeded, met, nearly met, or did not meet the achievement standard. It also reports results for specific content areas, such as reading, writing, problem solving, and communicating mathematical reasoning. For this analysis, I use the mean scale score as both the predictor and response because it provides a continuous measure of overall subject-level achievement and preserves more information than grouping students into achievement categories.

Because the dataset contains aggregate school-level results rather than individual student records, the analysis describes differences between schools and cannot be used to draw conclusions about individual students. In addition, schools may differ in enrollment, student demographics, participation rates, and the number of students with valid scores.

To examine the association between Math and ELA/Literacy performance, I first plotted the two mean scale scores for each school. Figure 1 shows the resulting scatter plot.

Overall, there appears to be a positive correlation, and surprisingly only a small percentage of schools' mean scores meet the Standard Met threshold for Grade 11 in 2024 (ELA mean scale score $\geq$ 2583 and Math mean scale score $\geq$ 2628) (California Department of Education 2024c). Table 1 shows summary statistics of the data.

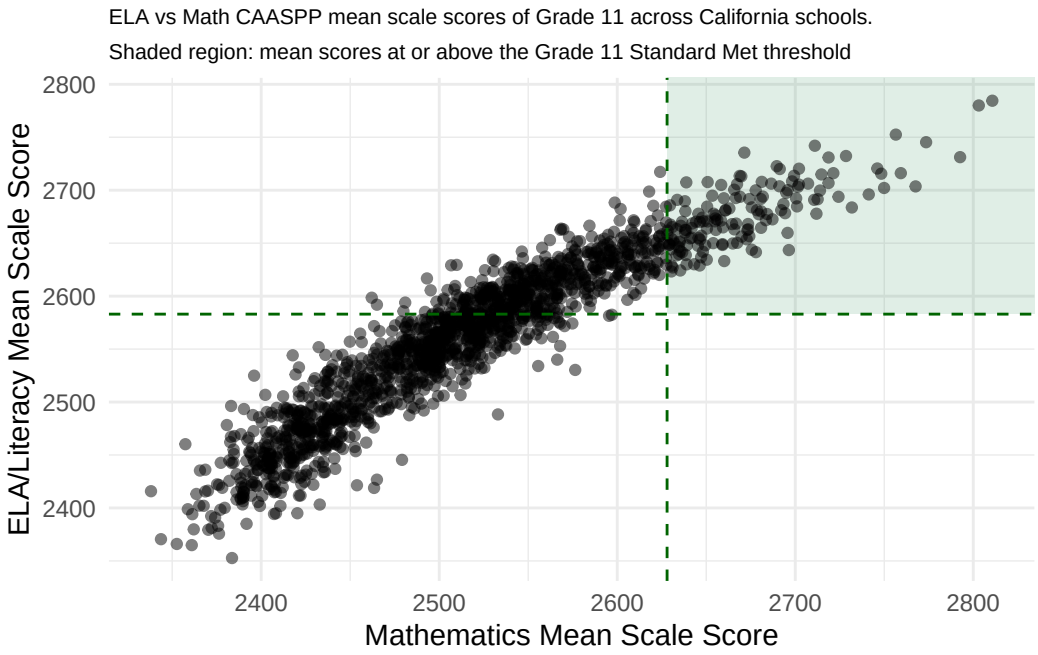


Figure 1: ELA vs Math CAASPP mean scale scores of Grade 11 across California schools.

Table 1: Summary statistics of the data

Variable	Min.	1st Qu.	Median	Mean	SD	3rd Qu.	Max.
ELA students tested	11.0	47.25	171.50	233.31	202.20	398.00	1030.0
Math students tested	11.0	47.25	171.50	232.80	201.91	397.00	1017.0
ELA mean scale score	2352.7	2505.60	2568.25	2559.76	74.62	2612.57	2784.5
Math mean scale score	2338.1	2452.52	2513.70	2515.27	79.82	2563.25	2810.7

The data include an average of approximately 233 students tested per school in ELA and 233 students tested per school in Math. However, the number of students tested varies considerably, ranging from 11 to 1,030 for ELA and from 11 to 1,017 for Math. ELA mean scale scores ranged from 2352.7 to 2784.5 with an average of 2559.76, compared with 2338.1 to 2810.7 and an average of 2515.27 for Math mean scale scores. The standard deviations of the school mean scale scores were 74.62 points for ELA and 79.82 points for Math, indicating a moderate amount of variation in average performance across schools. In contrast, the standard deviations for the number of students tested were approximately 202 students, nearly as large as the corresponding means, reflecting substantial differences in school size.

IV. Methods

Simple Linear Regression Model

To examine the association between mathematics and ELA/Literacy achievement, I used a simple linear regression model. The response variable is the mean Grade 11 ELA/Literacy scale score, and the explanatory variable is the mean Grade 11 mathematics scale score for the same school. The model is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

where Y_i is the mean ELA/Literacy scale score for school i, X_i is the mean mathematics scale score for school i, β_0 is the intercept, β_1 is the expected change in mean ELA/Literacy score associated with a one-point increase in mean mathematics score, and ε_i is the random error. Reversing the predictor and response would not change the existence of an association, but it would produce a different regression equation and predictions.

Hypothesis Test

To determine whether school mean Math scores are linearly associated with school mean ELA scores, we can conduct a hypothesis test at a significance level of $\alpha = 0.05$:

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_a : \beta_1 \neq 0.$$

The test statistic is

$$t = \frac{\hat{\beta}_1 - 0}{\widehat{\text{SE}}(\hat{\beta}_1)},$$

which follows a t distribution with $n - 2$ degrees of freedom when H_0 is true and the model assumptions hold. We reject H_0 if the p-value is less than 0.05.

Model Assumptions and Validation

1. Validity

CAASPP mean scale scores directly measure Grade 11 achievement in mathematics and ELA/Literacy, which are the two quantities in the research question. Linear regression is useful because it quantifies the expected change in ELA/Literacy mean scale score associated with a change in mathematics mean scale score.

2. Representativeness

The representativeness assumption is reasonable when the target population is defined as California schools with valid Grade 11 CAASPP results during the 2023-24 school year. Generalization to individual students, schools outside California, or other years can be made but needs careful interpretation.

3. Additivity and Linearity

It is reasonable to assume that Math mean scale score and ELA mean scale score have a linear relationship because the scatter plot of the data shows a roughly linear trend.

4. Independence of Errors

Schools within the same district may share policies, resources, and student characteristics, so their errors may be correlated. Therefore, complete independence of the school-level errors cannot be guaranteed. This potential dependence is treated as a limitation of the analysis.

5. Equal Variance of Errors

The residuals should have approximately constant variance across the range of fitted values. This can be evaluated using a residuals versus fitted-values plot after fitting the model.

6. Normality of Errors

This assumption is relevant primarily for confidence intervals and hypothesis tests. We can use a normal Q-Q plot to assess whether the residuals are approximately normally distributed after fitting the model.

Software

I conducted the analysis in R using RStudio and prepared the report with Quarto. The data were cleaned and transformed using the `tidyverse` packages, and the regression model was fitted using the `lm()` function. Figures were created using `ggplot2`, tables were created using `knitr`, and the final report was rendered as a PDF using Quarto (R Core Team 2025; Wickham et al. 2019; Xie 2015; Posit, PBC 2025).

V. Results

Table 2 summarizes the fitted model.

Table 2: Simple linear regression of school mean ELA scale score on mean Math scale score (n = 1630 schools, $R^2 = 0.879$, residual SE = 25.96).

Term	Estimate	Std. Error	t value	p-value
Intercept	355.4317	20.2810	17.53	< 0.001
Math mean scale score	0.8764	0.0081	108.74	< 0.001

The fitted simple linear regression model examined the association between ELA and Math mean scale score of schools across California. The estimated model was

$$\widehat{\text{ELA Mean Scale Score}} = 355.4 + 0.8764(\text{Math Mean Scale Score}).$$

The estimated regression slope was 0.8764, meaning that a one-point increase in a school's mean Math scale score is associated with an estimated 0.8764-point increase in its mean ELA scale score.

To test whether this association is statistically significant, I test $H_0 : \beta_1 = 0$ against $H_a : \beta_1 \neq 0$. From Table 2, the test statistic is $t = \hat{\beta}_1 / \widehat{\text{SE}}(\hat{\beta}_1) = 108.74$ with $n - 2 = 1628$ degrees of freedom, and the p-value is less than 0.001. Because the p-value is below $\alpha = 0.05$, we reject H_0 and conclude that there is strong evidence of a linear association between school mean ELA and Math scale scores.

The estimated intercept was 355.4, which is the predicted mean ELA score for a school with a mean Math score of zero. This value is far outside the observed data range, so the intercept is not very meaningful. The model had an R^2 value of 0.879, meaning that approximately 87.9% of the variation in school-level ELA scores was explained by the linear relationship with Math scores. Since the slope is positive, the correlation coefficient is $r = \sqrt{R^2} = \sqrt{0.879} \approx 0.938$, which indicates a strong positive correlation. Figure 2 shows the fitted regression line.

The upward trend indicates that schools with higher mathematics proficiency generally also tend to have higher ELA proficiency. The spread of points around the regression line is fairly even; however, toward higher Math mean scale scores, the points tend to fall below the line. Figure 3 shows the residuals plotted against the fitted values.

The residual plot suggests that the relationship is not perfectly linear and that the variance of the errors is not constant. In particular, the residuals show a curved pattern (upside-down U) and unequal spread across fitted values. The line over-predicts ELA for the lowest- and highest-scoring schools and under-predicts it for schools in the middle. This means the real relationship is concave: ELA rises quickly with Math among low-scoring schools and levels off among high-scoring ones. The residuals also spread from -100 to 90 for fitted values from 2500 to 2600, whereas for fitted values from 2600 to 2700 the spread is smaller, from -80 to 60. Figure 4 shows the normal Q-Q plot of the residuals.

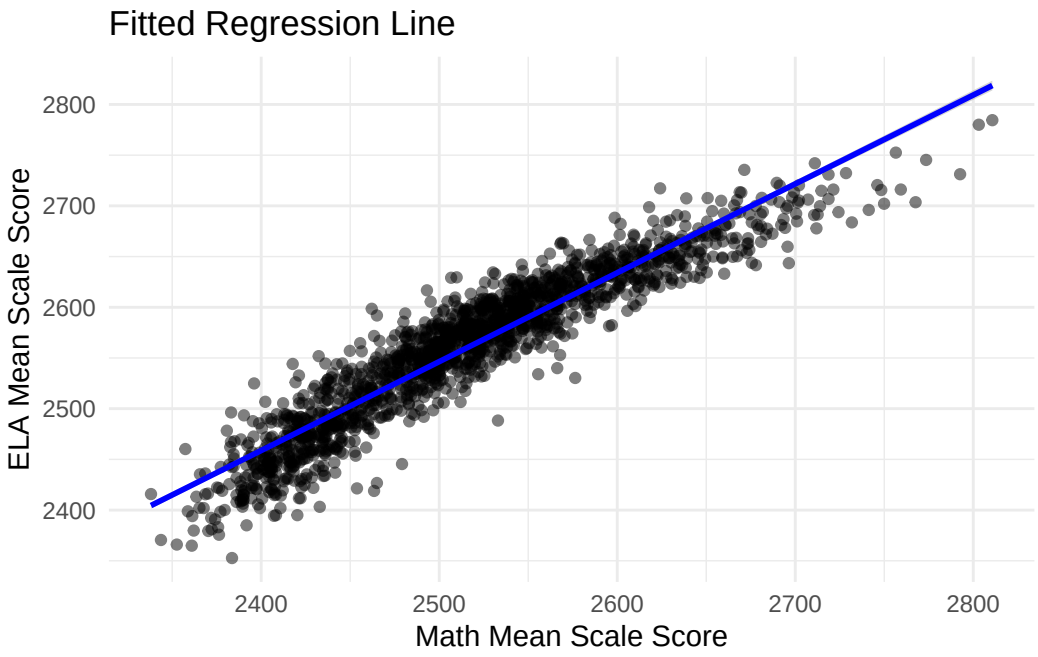


Figure 2: Fitted OLS line for ELA mean scale score on Math mean scale score

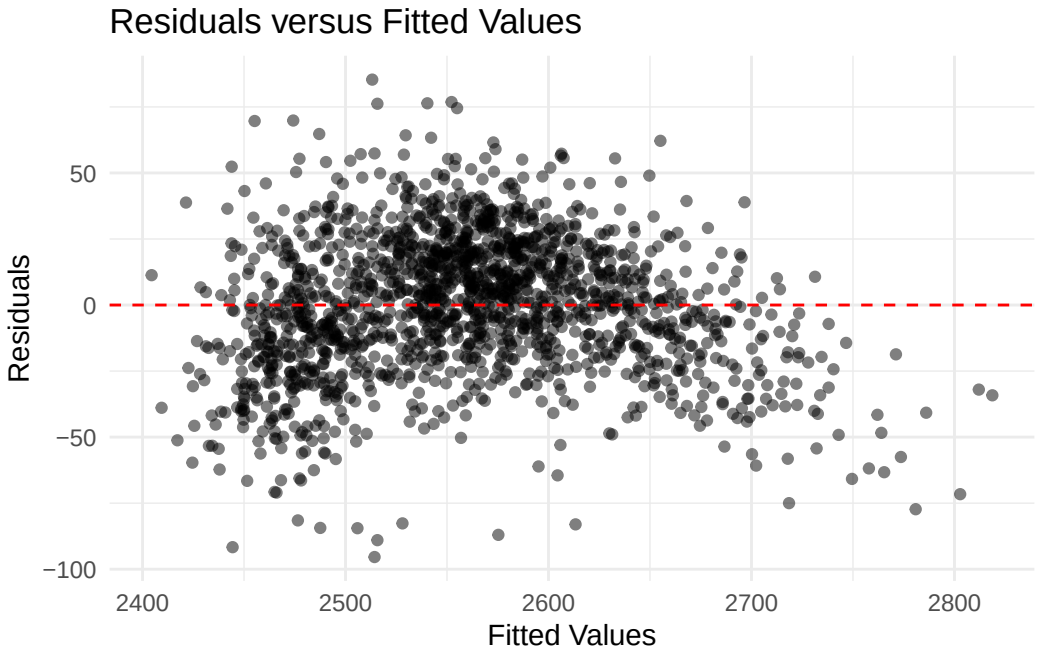


Figure 3: Residuals versus fitted values for the linear regression model

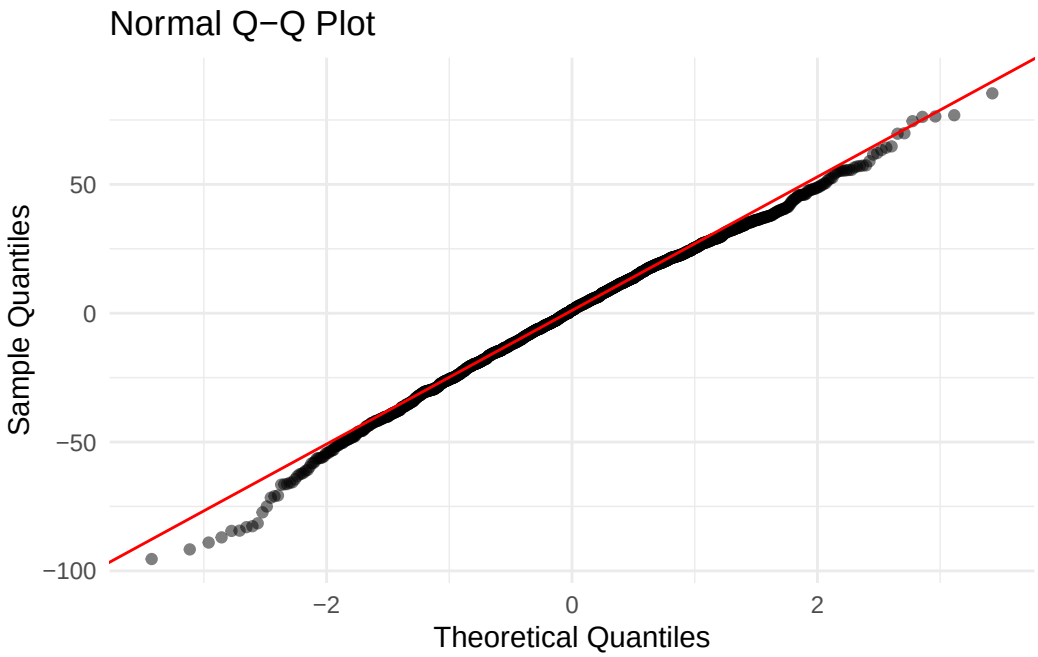


Figure 4: Normal Q-Q Plot

The normal Q–Q plot shows that most residuals lie close to the reference line, suggesting that the residuals are approximately normally distributed. There are some deviations in the tails, indicating a few unusually large residuals, but the normality assumption is still plausible.

The results answer the research question by showing a strong positive school-level association between mathematics and ELA proficiency in California, which contradicts our original belief. However, generalization to individual students should be made carefully. The association is likely to be stronger for lower-scoring schools and weaker for higher-scoring schools. I would expect a polynomial regression of degree two to improve the fit, but for the scope of this project, I only assess the performance of simple linear regression.

VI. Discussion

The regression showed a strong positive relationship between mathematics mean scale score and ELA/Literacy performance, with a high correlation coefficient of $r = 0.938$. A one-point increase in mathematics mean scale score is associated with an estimated 0.8764-point increase in ELA/Literacy mean scale score, and the hypothesis test showed that this association is statistically significant ($t = 108.74$, $p < 0.001$). The intercept was 355.4, which is the predicted mean ELA score when the mean Math score is 0; this is not very meaningful because it is far outside the observed range. The model explained approximately 87.9% of the variation in ELA/Literacy outcomes.

Several assumptions are not fully met. Because the residuals show a curved pattern, the slope describes the average association across all schools, and the model tends to overpredict mean ELA scores for the lowest- and highest-scoring schools. Schools are nested within districts, which means their errors may be correlated, and the spread of the residuals is not the same across all fitted values. These two violations do not bias the parameter estimates, but their standard errors are not reliable. As a result, the p-value is likely smaller than it should be, although with such a large t statistic the association would most likely still be significant. The association is also likely to be stronger for the lowest-scoring schools and weaker for the highest-scoring schools. Future work could try a polynomial model to better fit the curve.

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